CASE WESTERN RESERVE UNIVERSITY

LGBT Center Library Catalog

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CASE WESTERN RESERVE UNIVERSITY Division of Student Affairs
Lesbian Gay Bisexual Transgender Center

Introduction:

We created an online library catalog system for the Case LGBT Center, which currently manages checkouts and returns for its library manually with a spreadsheet. We sought to make the process of managing the LGBT Center library faster and easier for everyone involved, as well as creating a way to browse the contents of the LGBT Center library without necessarily having to be there physically.

Design Principles:

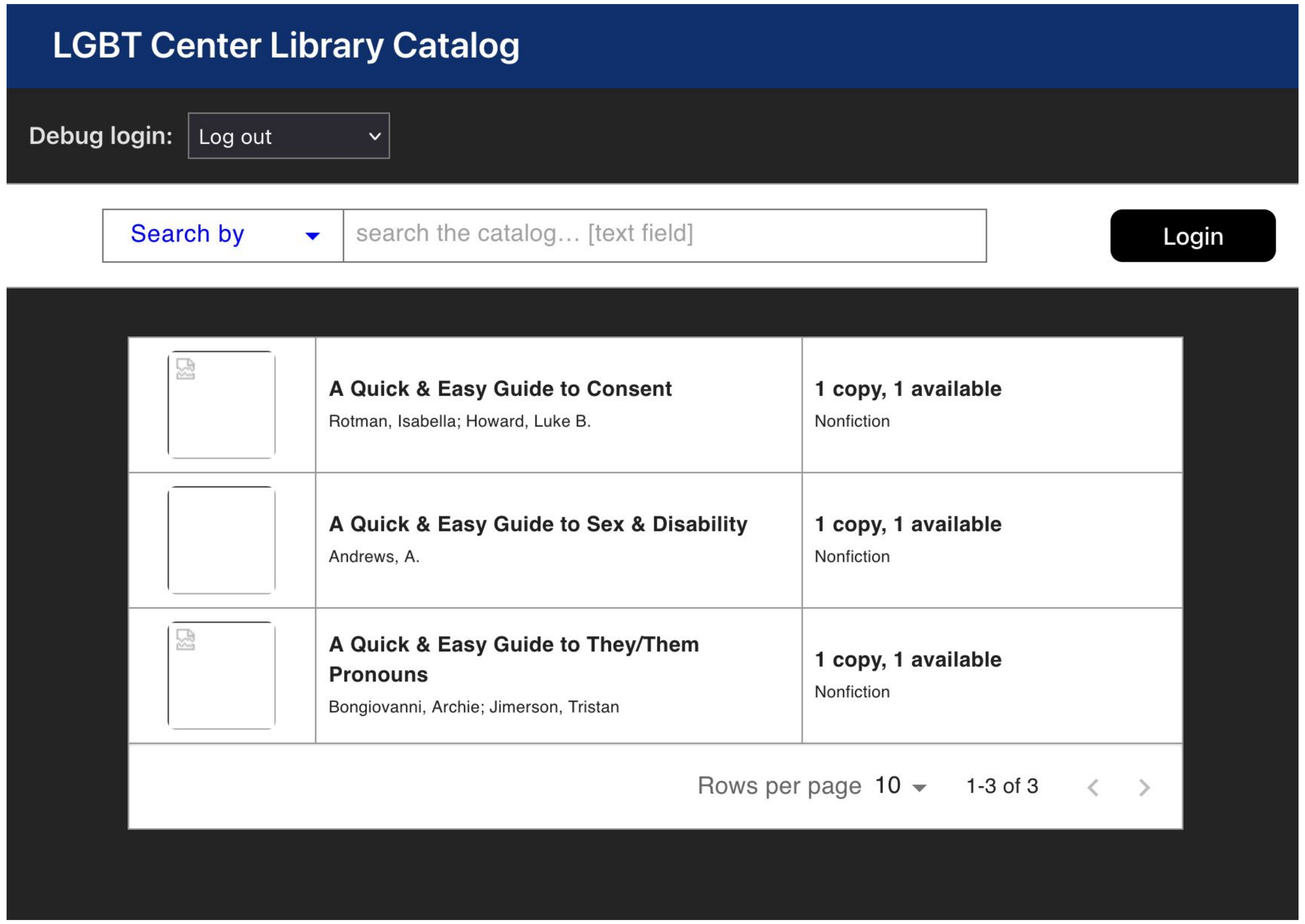
The web service was designed to preserve user privacy. Users are saved according to their caseIDs and once they return all their items, they no longer exist in the database.
In contrast, the library items are robustly cataloged with their genre, publisher, version, and copies notated.

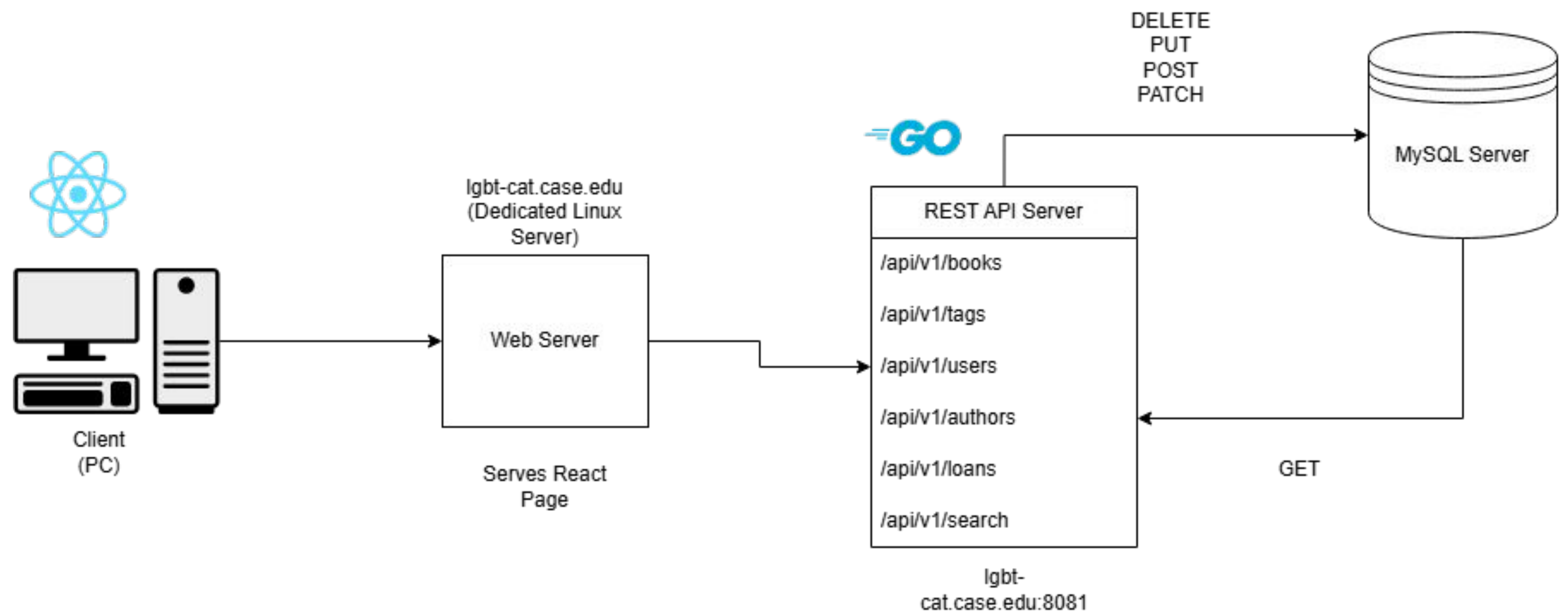
Design Implementation:

Frontend: Vite and React used to design and implement the webpage
Backend: Golang used to create the backend framework
Database: MySQL used to manage the database

User Authentication:

Users can log into the LGBT Center library system using their Case IDs, via the CAS single sign-on system (3rd party integration with Duo).





User Permissions:

There are three authentication levels users are grouped into: patrons, staff, and admin. Patrons can check out and renew books, staff can manage books in the catalog, and admin can manage permissions for individual accounts. Students will be assigned patron roles, LGBT Center staff will get staff permissions, and LGBT Center pro staff will get admin accounts. All site visitors can browse the catalog without logging in.

Testing:

We utilized integration testing using temp Docker containers for MySQL instances to test that the server and database can successfully communicate with each other as intended.

Future Work:

The system will need to be maintained to ensure service can continue to be provided to patrons of the LGBT Center. We may also plan to add supplemental features, such as sending email notifications to students confirming to them that an item has been checked out and reminding them of upcoming due dates on items they currently have checked out. We are also considering creating a CI/CD pipeline for an automated build + testing + deployment process to our dedicated Linux server, and shifting towards a microservice-oriented architecture (API Gateway).